

Radiology Impression Parser

Parsing multilingual radiology impressions into structured organ-level findings.

- Negation-aware Rules
- Clinical NLP
- TF-IDF + LogReg
- Fine-tuned IndoBERT

Synthetic data. Every patient, result, and label is fabricated by the Dashlabs generator — no real person or finding appears. The work demonstrates the *method*; results are illustrative.

01 · BUSINESS PROBLEM

Structure free-text impressions — and generalize honestly.

Thirteen synthetic diagnostic-lab clients store **679 imaging impressions** (USG abdomen and chest X-ray) as free text — Bahasa Indonesia mixed with Latin and English organ terms, labeled normal/abnormal via `rad_label` (~60% normal). On this *closed* synthetic vocabulary every method looks near-perfect — so the project measures the honest question: **generalization to a finding whose vocabulary is unseen**, where a hand-written rule set and a trained model part ways.

Multilingual

PROBLEM TYPE

Clinical NLP

3 methods

COMPARED

rules · TF-IDF · IndoBERT

0 → 93%

UNSEEN-FINDING
RECALL

rules → IndoBERT

60%

NORMAL RATE

405 / 679

02 · DATA SOURCES

Impressions live in `word_value`.

TABLE	ROLE	KEY COLUMNS
<code>patient_service_results</code>	PRIMARY	<code>word_value</code> — free-text impression · <code>service_name</code> — Ultrasound / USG Abdomen / Rontgen Dada · <code>rad_label</code> — human ground truth
<code>patient_services</code>	context	<code>service_name</code> · <code>status</code> · <code>created_at</code>

03 · METHODOLOGY

Rules built around negation — then stress-tested for generalization.

- 01

Corpus analysis

Catalog phrasings, cross-language synonyms, and negation forms (tidak tampak, no, unremarkable).
- 02

Normalize + clause / negation rules

Split into clauses; flag abnormal only on an un-negated finding cue ("abnormal wins").
- 03

Train real models

Genuinely train TF-IDF + LogReg and fine-tune IndoBERT on the same leakage-free split (seed 42).
- 04

Vocabulary-held-out generalization test

Hold effusion + stones out of BOTH the rule vocabulary and the models' training data; score all three on the now-unseen cues.

Structured findings extracted — and generalization measured honestly.

Computed on the 679 labeled impressions. In-distribution every method saturates on the closed synthetic vocabulary; the discriminating results are the two generalization tests below.

0 → 93%

UNSEEN-FINDING
RECALL

sole-signal: rules →
IndoBERT

~100%

IN-DIST ACCURACY
closed vocab (secondary)

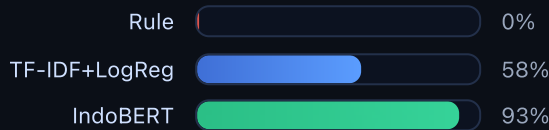
679

IMPRESSIONS
607 distinct

6

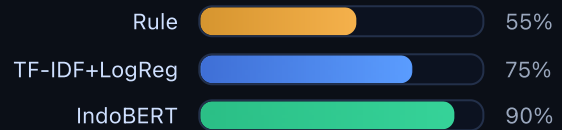
ORGANS TRACKED
parsed

Recall on truly-unseen findings (sole-signal held-out)



effusion + stones held out of vocabulary AND training (40 sole-signal abnormalities). Rules can't match unseen cues; the transformer generalizes from context.

Generalization to novel phrasings — OOD accuracy



20 hand-written impressions whose vocabulary sits outside the corpus — the harder, complementary test.

Organ mentions across impressions



Rules can't generalize to unseen vocabulary — learned models can

Hold effusion + stones out of everything and the rule system scores **0%** recall where an unseen cue is the only signal — it cannot match a word it has never seen. TF-IDF recovers 58% from context; the fine-tuned IndoBERT reaches **93%**. That contrast, not the 100%, is the story.

Negation is the biggest lever (in-distribution)

A naïve "mentions *normal*" heuristic scored 63.8% and missed ~81% of abnormalities — "cor membesar" sits beside the word "normal". Clause-level negation is what lifts the rules to 100% on the closed vocabulary.

Synonyms bridge languages

One organ, many surface forms — gallbladder / vesica fellea / batu empedu; ginjal / kidney / renal — handled by a synonym map so the same rules span Bahasa, Latin, and English in one sentence.

Liver and lungs lead mentions

Hepar (325) and pulmo (314) are named most often — consistent with abdominal USG plus chest X-ray.

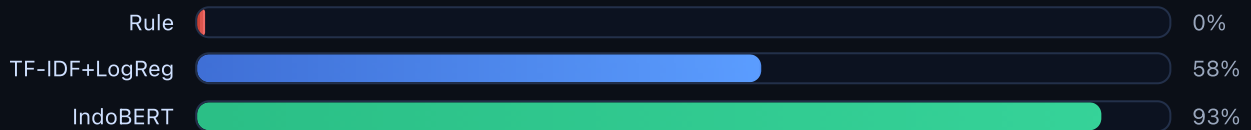
Is the 100% too good to be true?

Yes — it is correct, but only on a closed vocabulary. Here is the honest generalization number, measured.

Reading note — the 100% is on a closed vocabulary

The data is synthetic from a fixed phrase set, so the held-out test shares its vocabulary with training. This is **not** classic overfitting (no train/test gap; the rules have no learned weights to memorize) — its only blind spot is an **unseen finding term**. To measure that blind spot fairly, two whole finding families — **effusion + stones** (10 cues) — are removed from **both** the rule vocabulary **and** the trained models' data, then all three are scored on the 274 impressions that use them (40 where the unseen cue is the *only* signal).

Recall on sole-signal unseen findings — the honest generalization number



The rule system catches **none** of these — by construction it can only match cues it was written for. TF-IDF recovers 58% from surrounding context, and the fine-tuned IndoBERT **93%**. Note that *overall accuracy* barely moves when one finding is unseen (such impressions are rare in the set) — which is exactly why accuracy hides this failure and recall on the unseen finding is the number to report.

Three methods, two axes

MODEL	IN-DIST ACC	OOD ACC	HELD-OUT ACC	SOLE-SIGNAL RECALL
Rule (regex + negation)	100%	55%	69%	0%
TF-IDF + LogReg	91%	75%	78%	58%
IndoBERT (fine-tuned)	99%	90%	89%	93%

Rules win in-distribution; learned models win on unseen findings — the tradeoff, made explicit.

06 · PRODUCTION PATH

Widen the held-out sweep

More finding families and an XLM-R comparison to map where the generalization gap opens.

Radiologist-labeled real reports

The only fully honest generalization figure needs a sample of real, human-labeled impressions.

Per-organ gold labels

Hand-label organ findings to score the structured extraction directly, not just the overall flag.

PACS integration & trends

Link findings to images; track organ findings per patient across visits.